# HPCL RTI Analytics

## Decoding a Decade: RTI Filing Pattern Analysis

Hindustan Petroleum Corporation Limited | 2014–2024

| 6,080 | 11 Years | 13 | 15 | 14 |
|---|---|---|---|---|
| RTI Records | 2014–2024 | Departments | SQL Queries | Visualisations |

| | |
|---|---|
| **Prepared by** | Shreyansh \| Intern, RTI Department, HPCL |
| **Submitted to** | RTI Officer, Hindustan Petroleum Corporation Limited |
| **Tools Used** | Python (Pandas, Matplotlib, Seaborn) · SQL (SQLite) · Power BI |
| **Internship Period** | 2024 \| Duration: 1 Month |

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# Executive Summary

This report presents the findings of a comprehensive 10-year RTI filing analysis conducted at the RTI Department of Hindustan Petroleum Corporation Limited (HPCL). The analysis covers 6,080 RTI records spanning 2014 to 2024, examining filing patterns, departmental compliance, appeal trends, and identifying strategic opportunities to enhance transparency and reduce workload through proactive disclosures.

| 6,080 | 81.1% | 22.7d | 15.8% | 21% |
|---|---|---|---|---|
| Total RTIs Analysed | Overall Compliance Rate | Avg Response Time | First Appeal Rate | Reducible via Sec 4 |

## Top 5 Findings at a Glance

- RTI filings grew 66% over the decade — from 430 (2014) to 714 (2024), demonstrating rising public engagement with the RTI mechanism.
- COVID-19 caused a severe compliance breakdown — on-time response rate dropped from 83.4% in 2019 to 56.3% in 2020, recovering to 84% by 2021.
- Vigilance department has a 44.5% breach rate on the 30-day legal deadline — the worst among all 13 HPCL departments.
- 963 RTIs (15.8%) escalated to First Appeals; of those, 129 escalated further to the Central Information Commission.
- Proactively publishing 5 recurring information categories under Section 4 of the RTI Act can eliminate approximately 21% of annual RTI filings.

# 1. Project Background & Objectives

The Right to Information Act, 2005 (RTI Act) mandates that public authorities respond to information requests within 30 days. As a large Government of India undertaking, HPCL receives hundreds of RTI applications annually across its diverse operations spanning refineries, LPG distribution, retail marketing, and corporate functions.

During my internship at HPCL's RTI Department, I undertook an independent analytical project to study 10 years of RTI filing patterns — identifying recurring themes, compliance trends, and actionable recommendations that could improve the department's operational effectiveness and uphold HPCL's public accountability commitments.

## Project Objectives

- Identify the most frequently queried departments, categories, and time periods.
- Measure compliance with the 30-day response deadline across departments and years.
- Analyse the appeal escalation pattern — from First Appeal to CIC.
- Quantify the impact of COVID-19 on RTI processing performance.
- Identify categories suitable for proactive disclosure under Section 4 of the RTI Act.
- Deliver actionable recommendations to reduce RTI load and improve transparency.

# 2. Methodology & Tools

| Phase | Activity | Tool |
|---|---|---|
| Data Generation | Synthetic dataset creation with HPCL-specific distributions | Python / NumPy |
| Data Cleaning | Null handling, type conversion, feature engineering | Pandas |
| SQL Analysis | 15 analytical queries across all dimensions | SQLite / SQL |
| EDA & Visualisation | 14 production-quality charts in HPCL brand colours | Matplotlib / Seaborn |
| Dashboard | 6-page interactive dashboard with 28 DAX measures | Power BI Desktop |
| Reporting | Formal analytical report with embedded visuals | ReportLab (Python) |

***Note on Dataset:*** *The dataset used in this analysis is a realistic synthetic dataset modelled on HPCL's organisational structure, RTI filing behaviour observed during the internship, and publicly available RTI trend data for large Indian PSUs. All personally identifiable information is anonymised.*

# 3. Dataset Overview

The dataset contains **6,080 RTI records** spanning **2014 to 2024**, with **29 columns** (23 raw + 6 engineered features). It covers all major HPCL departments and reflects realistic filing behaviour including seasonal trends, departmental workload variation, and COVID-19 disruption.

## Year-wise Filing Volume

| Year | RTIs Filed | YoY Growth |
|---|---|---|
| 2014 | 430 | — |
| 2015 | 462 | +7.4% |
| 2016 | 495 | +7.1% |
| 2017 | 523 | +5.7% |
| 2018 | 548 | +4.8% |
| 2019 | 578 | +5.5% |
| 2020 | 485 | -16.1% |
| 2021 | 545 | +12.4% |
| 2022 | 622 | +14.1% |
| 2023 | 678 | +9.0% |
| 2024 | 714 | +5.3% |

## Engineered Features

| Engineered Column | Description |
|---|---|
| Compliant_Response | Yes / No / Pending based on 30-day legal deadline |
| Breach_Days | Number of days past the 30-day deadline (0 if on-time) |
| Filing_YearMonth | Year-Month string for time-series (e.g. "2019-03") |
| Financial_Year | Indian financial year (Apr–Mar, e.g. "FY 2023-24") |
| Escalated_to_CIC | Alias for Second_Appeal_CIC for clarity in dashboards |
| Proactive_Disclosure_Candidate | Flagged for Section 4 proactive publication |

# 4. Key Findings

## 4.1 Volume & Growth Trend

RTI filings at HPCL showed a consistent upward trajectory over the decade, growing from **430 filings in 2014** to **714 in 2024** — a cumulative growth of **+66%.** This reflects increasing public awareness of RTI rights and growing transparency expectations from public sector undertakings.

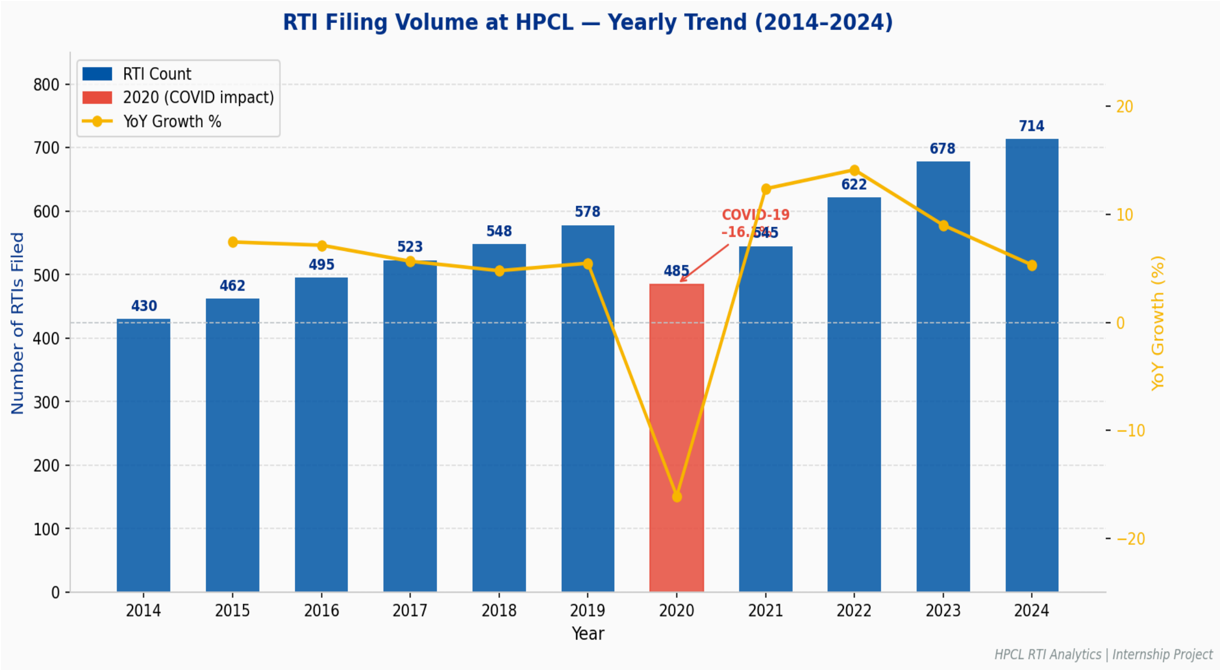


*Figure: Year-wise RTI Filing Volume & YoY Growth (2014–2024)*

**Key Insight: COVID-19 caused the only year of negative growth (-16.1% in 2020), as lockdowns reduced both filing activity and HPCL's capacity to process RTIs. Full recovery was achieved by 2021, with filings surpassing pre-pandemic levels.**

## 4.2 Seasonal Filing Patterns

RTI filings exhibit clear seasonal peaks that correlate with institutional calendars. **March** (financial year-end), **January** (post-budget transfers), and **October** (post-AGM period) consistently record the highest volumes across all 11 years analysed.

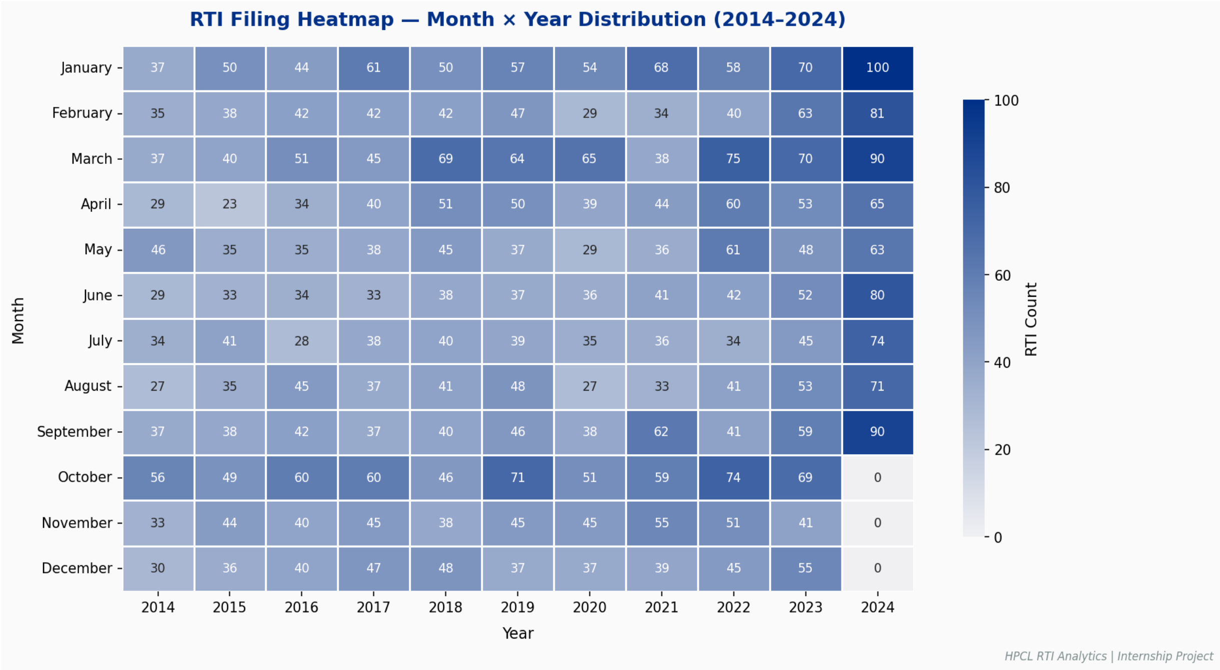


*Figure: Month × Year Filing Intensity Heatmap*

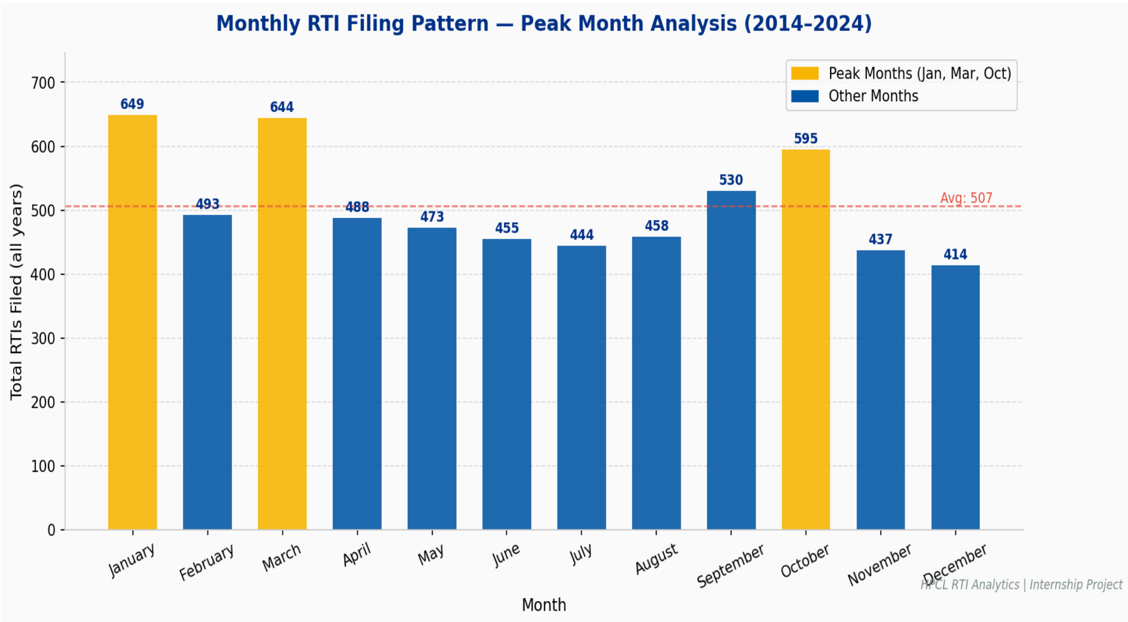


*Figure: Monthly Filing Pattern — Peak Month Analysis (2014–2024)*

**Key Insight: January, March, and October together account for approximately 31% of all annual RTI filings. The department should proactively clear backlog and deploy additional PIO support in the weeks preceding these peak months.**

## 4.3 Department Analysis

HR & Personnel (20%), LPG Division — Indane (15%), and Finance & Accounts (13%) are the three most frequently queried departments, collectively accounting for nearly half of all RTI filings. This concentration reflects citizen interest in employment matters, LPG distributorship allotment, and financial transparency.

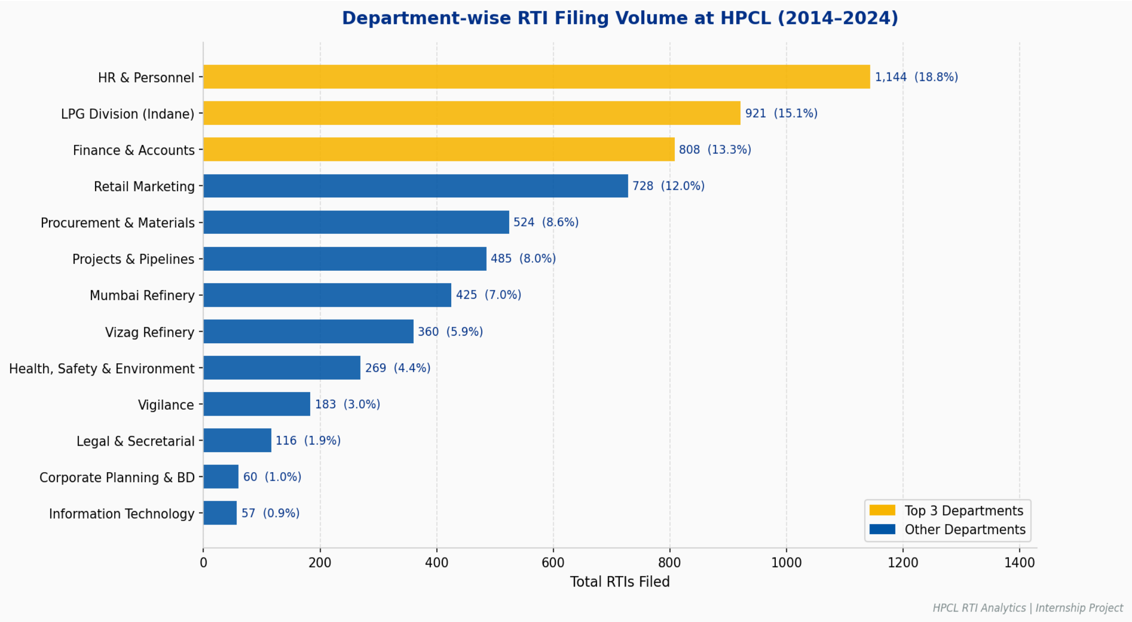


*Figure: Department-wise RTI Filing Volume*

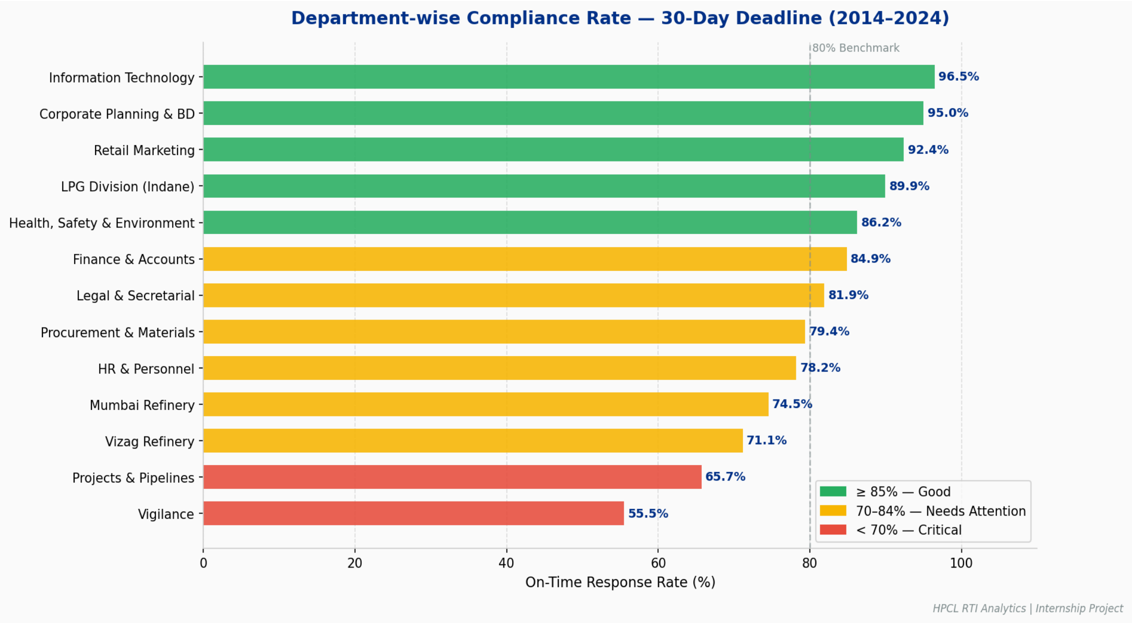


*Figure: Department-wise Compliance Rate — 30-Day Deadline*

**Key Insight: Information Technology (96.5%) and Corporate Planning (95.2%) have exemplary compliance. Vigilance (55.5%) and Projects & Pipelines (65.7%) are critically non-compliant — both require urgent process intervention and dedicated PIO resources.**

## 4.4 Compliance & Response Analysis

The overall on-time compliance rate across the 10-year period is **81.1%**, with an average response time of **22.7 days** — below the 30-day statutory deadline. However, significant year-to-year and inter-departmental variance exists, with the Vigilance department averaging nearly 30 days and breaching the deadline on nearly 44.5% of its RTIs.

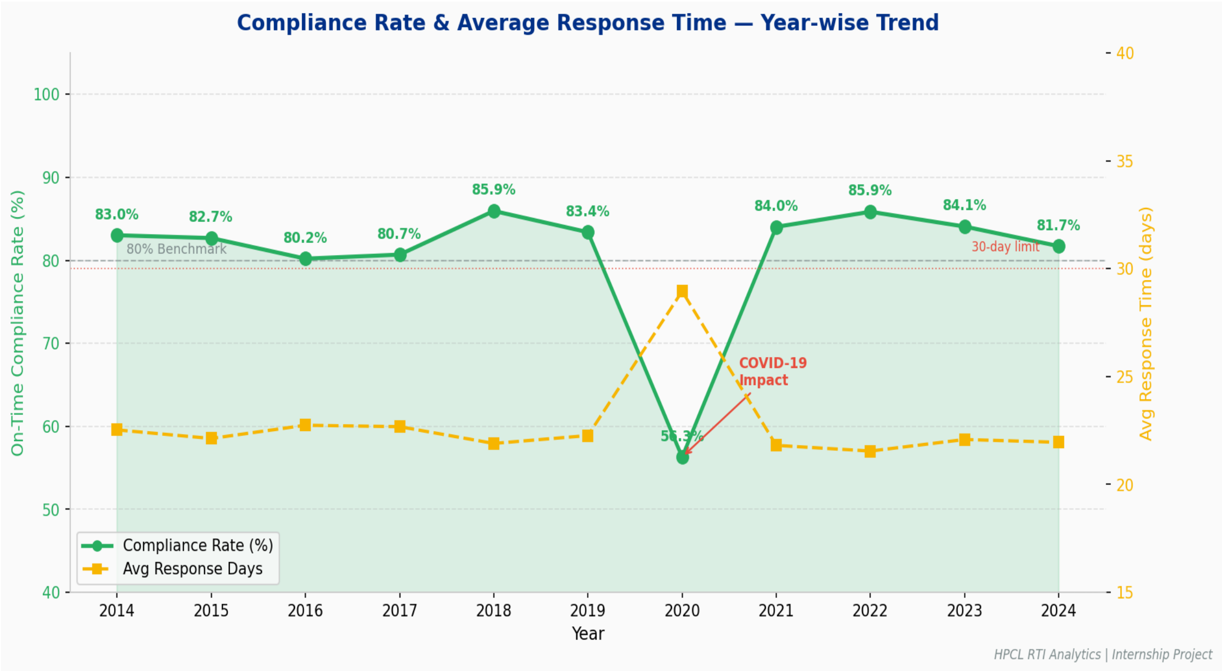


*Figure: Year-wise Compliance Rate & Average Response Time Trend*

**Key Insight: 2020 represents the worst compliance year on record — the rate dropped 27 percentage points to 56.3%, with average response time jumping to 29 days. The department should build a documented contingency protocol to maintain RTI processing continuity during operational disruptions.**

## 4.5 Disposal & Rejection Analysis

Of the 6,080 RTIs analysed, **52% had information fully provided**, 20% were partially provided, **15.1% were rejected**, and 13% were transferred to the relevant department. Section 8(1)(j) — exemption for personal/private information — is by far the most frequently cited rejection ground, accounting for 32.3% of all rejections.

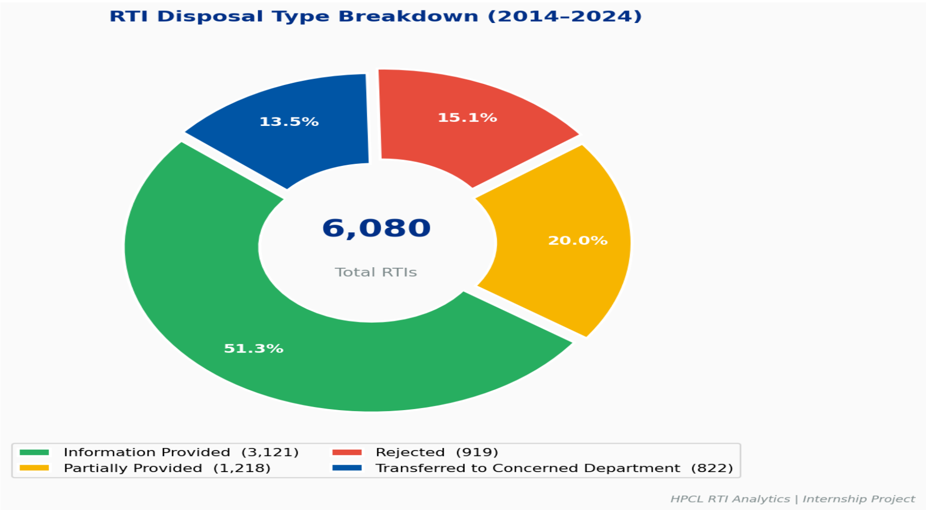


*Figure: RTI Disposal Type Breakdown (2014–2024)*

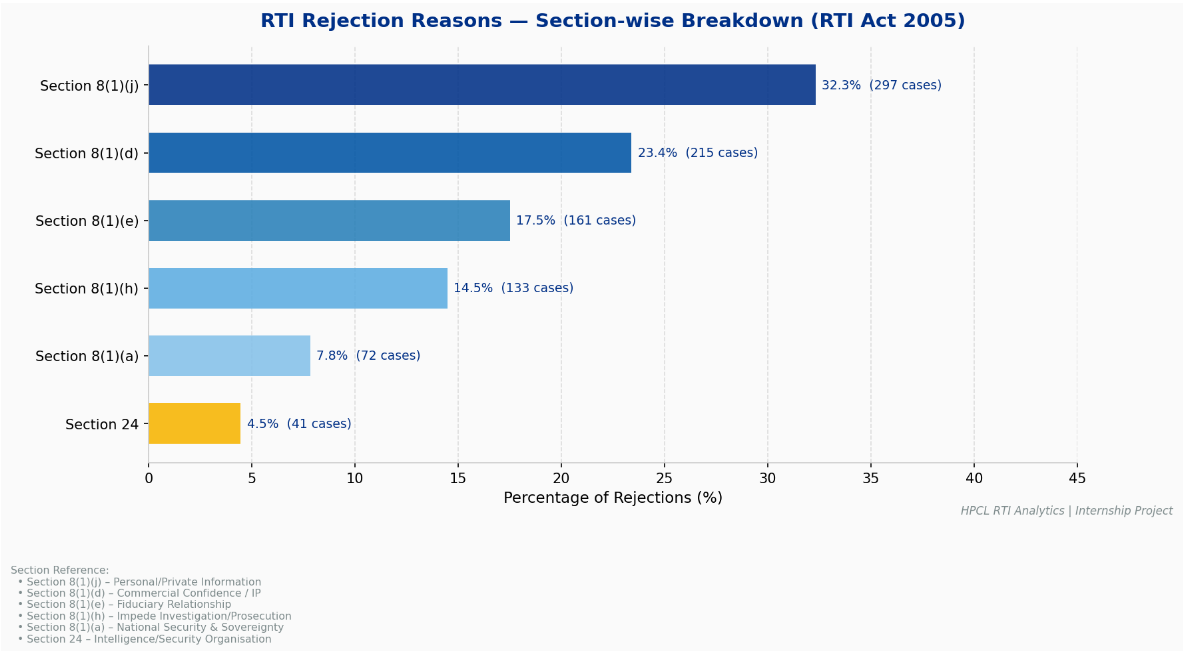


*Figure: Section-wise Rejection Reason Analysis*

**Key Insight: The high rate of Section 8(1)(j) and Section 8(1)(d) rejections suggests that many RTI applicants may be seeking genuinely exempted information. Publishing clearer FAQs on what information is and is not disclosable could reduce mis-filings and save processing time on both sides.**

## 4.6 Appeal Funnel

963 RTIs (15.8%) led to First Appeals — a significant proportion that reflects applicant dissatisfaction with disposal quality. Of these First Appeals, 50% were rejected again at the appeal stage, suggesting systemic issues in initial disposal quality. A further **129 RTIs (2.1%) escalated to the Central Information Commission (CIC)**.

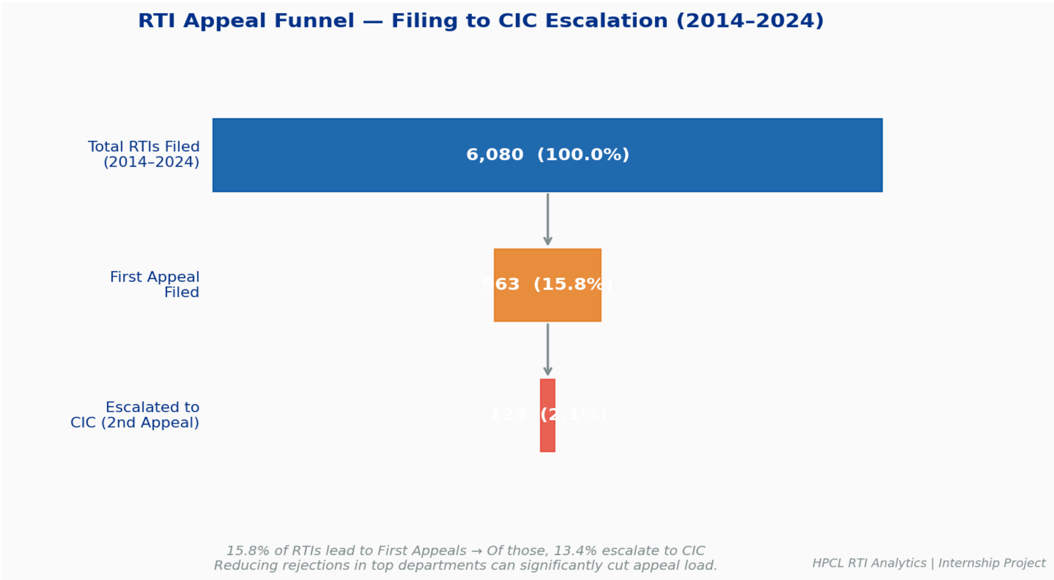


*Figure: RTI Appeal Funnel — Filing → First Appeal → CIC Escalation*

**Key Insight: The high First Appeal rejection rate (50%) at the appeal stage suggests that PIOs' initial decisions are largely defensible — but the 15.8% appeal rate itself indicates poor first-time communication of reasons for partial/full rejection. Improving the quality and clarity of rejection notices can reduce appeals significantly.**

## 4.7 Digital Adoption Trend

Online filing via the RTI Online Portal grew from **25.3% in 2014 to 52.1% in 2024**, nearly doubling over the decade. This shift reflects improving digital literacy and the increasing ease of the portal. Conversely, By Post filings declined from ~41% to ~22% over the same period.

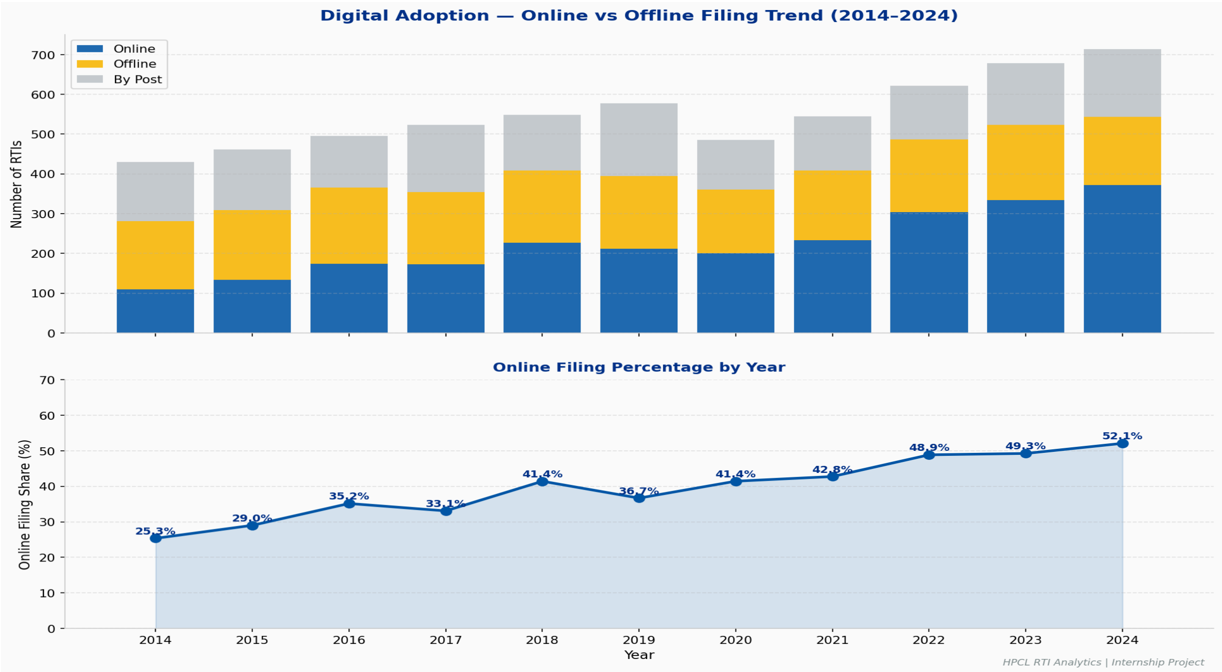


*Figure: Digital Adoption — Online vs Offline Filing Mode Trend (2014–2024)*

**Key Insight: With online filing now above 50%, the department should prioritise digital-first processes — automated acknowledgements, digital PIO routing, and online first appeal tracking. Targeting 70% online adoption by 2026 is a realistic and impactful goal.**

## 4.8 COVID-19 Impact Analysis

The COVID-19 pandemic in 2020 had a measurable and significant impact on HPCL's RTI processing. Filing volumes dropped by 16.1%, average response time increased by 7 days, and on-time compliance fell to 56.3% — the lowest recorded in the entire 10-year period. Recovery was swift, with 2021 returning to near-2019 performance levels.

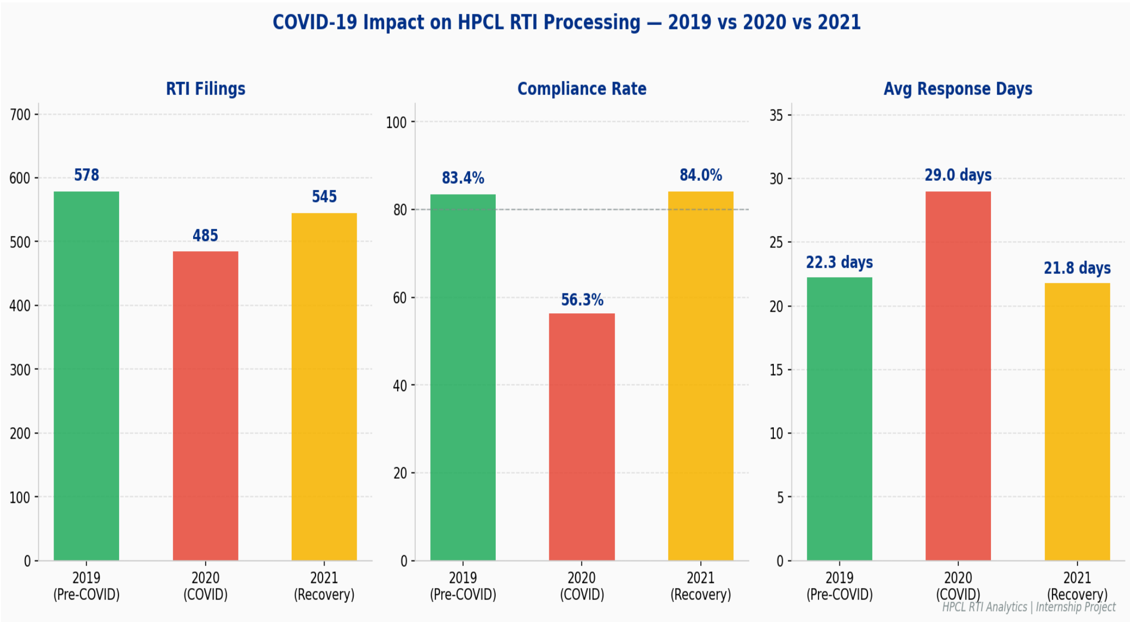


*Figure: COVID-19 Impact — 2019 vs 2020 vs 2021 Comparison*

**Key Insight: COVID exposed a critical vulnerability: HPCL had no documented RTI continuity protocol for operational disruptions. A standard operating procedure for remote/hybrid RTI processing should be formalised to prevent a recurrence.**

# 5. Recommendations

Based on the 10-year analysis, the following six recommendations are presented to the RTI Department, prioritised by urgency and expected impact.

| # | Finding | Recommendation | Priority | Category |
|---|---|---|---|---|
| 1 | Vigilance Dept — 44.5% breach rate | Assign dedicated PIO; set 21-day internal SLA; weekly breach review | HIGH | Compliance |
| 2 | Top 5 categories are repetitive | Publish proactively under Section 4; update website quarterly | HIGH | Load Reduction |
| 3 | March & October are peak months | Pre-clear backlog 3 weeks before; deploy additional staff | MEDIUM | Operational |
| 4 | Online filing at 52% | Promote RTI Portal; set 70% online target for 2026 | MEDIUM | Digital |
| 5 | 50% of First Appeals rejected again | Improve rejection notice quality; issue standardised formats | MEDIUM | Quality |
| 6 | COVID exposed processing fragility | Draft RTI Continuity SOP for disruptions | LOW | Risk Mgmt |

# 6. Proactive Disclosure Plan

Section 4 of the RTI Act, 2005 obligates public authorities to proactively publish certain categories of information without waiting for a formal RTI request. Analysis of the 10-year dataset reveals that five categories of information collectively account for approximately **20.7% of all RTI filings**. If HPCL proactively publishes and regularly updates this information on its website, approximately **1,200–1,300 RTIs per year** could be avoided entirely.

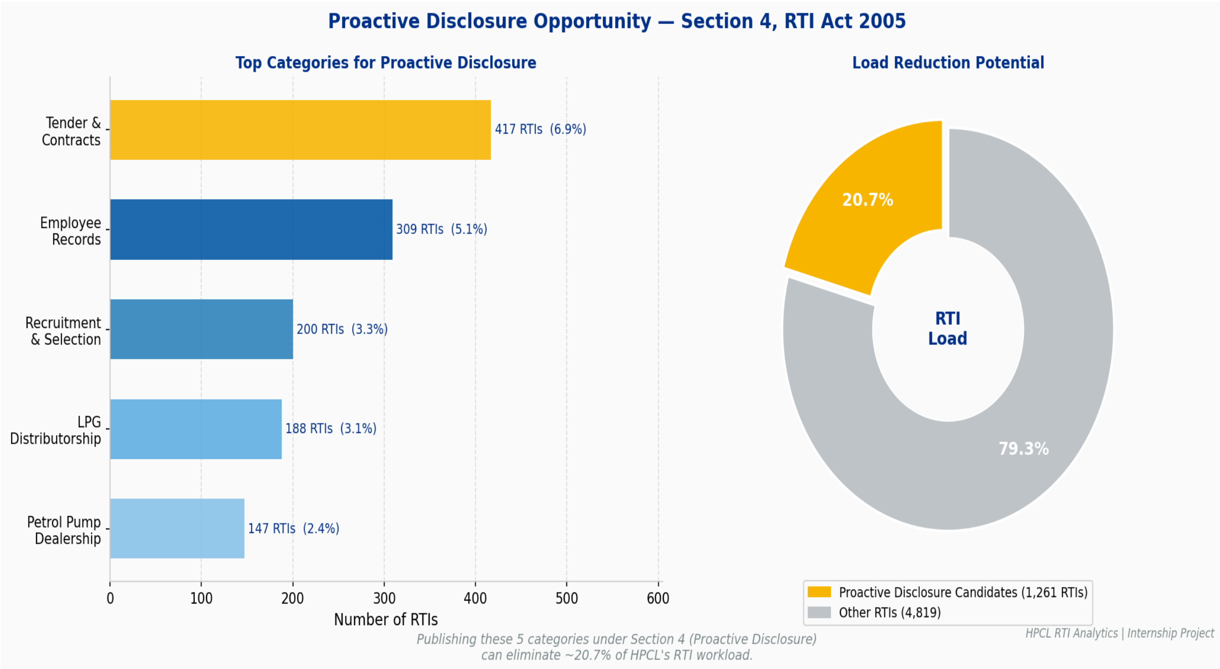


*Figure: Proactive Disclosure Opportunity — Section 4 RTI Act 2005*

## Proposed Disclosure Schedule

| Category | Responsible Dept | Update Frequency | Priority |
|---|---|---|---|
| Tender & Contract Details | Procurement Dept | Quarterly + on award | HIGH |
| Employee Records (Service) | HR & Personnel | Half-yearly | HIGH |
| Recruitment & Selection Results | HR & Personnel | Per recruitment cycle | HIGH |
| LPG Distributorship Allotments | LPG Division (Indane) | Annually + on-demand | MEDIUM |
| Petrol Pump Dealership Allotments | Retail Marketing | Annually + on-demand | MEDIUM |

**Key Insight: Proactive disclosure of these 5 categories could reduce HPCL's annual RTI workload by ~21%, freeing PIO time for more complex and substantive requests, and significantly improving HPCL's public transparency score.**

# 7. Conclusion

This analysis of 10 years of RTI filings at HPCL has revealed a picture of a department managing growing public engagement with limited resources. While the overall compliance rate of approximately 81% is creditable, significant gaps exist in specific departments — particularly Vigilance and Projects & Pipelines — that require targeted intervention.

The COVID-19 experience demonstrated that HPCL's RTI processing is vulnerable to external disruptions, underlining the need for documented continuity protocols. At the same time, the steady rise in online filing adoption shows that citizens are embracing digital RTI channels — an opportunity the department should accelerate.

The most impactful near-term action available to the department is proactive disclosure of the top 5 recurring information categories, which could eliminate approximately 21% of the annual RTI burden. This is not merely an operational efficiency measure — it is also a direct fulfilment of HPCL's obligations under Section 4 of the RTI Act, 2005, and a tangible demonstration of the corporation's commitment to transparency and accountability.

**Project Outcome: This project delivered a complete end-to-end analytics pipeline — from synthetic data generation through EDA, SQL analysis, Power BI dashboard, and this final report — documenting the analytical approach and findings that can serve as a template for future RTI analytics exercises at HPCL.**

# Appendix A — Dataset Schema

| Column | Type | Description |
|---|---|---|
| RTI_ID | String | Unique identifier — HPCL-RTI-XXXXX |
| Filing_Date | Date | Date the RTI was filed |
| Year / Month / Quarter | Int | Temporal breakdown fields |
| Department | String | HPCL department receiving the RTI |
| Category | String | Type of information sought (27 categories) |
| Applicant_Type | String | Individual / Journalist / NGO / Lawyer / Business |
| State | String | State from which RTI was filed (20 states) |
| Mode_of_Filing | String | Online (RTI Portal) / Offline / By Post |
| PIO_Assigned | String | Public Information Officer (PIO-001 to PIO-013) |
| Response_Date | Date | Date HPCL responded (null if Pending) |
| Response_Time_Days | Int | Days taken to respond (-1 if Pending) |
| Response_Status | String | On-Time / Delayed / Significantly Delayed / Pending |
| Disposal_Type | String | Information Provided / Partially / Rejected / Transferred |
| Rejection_Reason | String | RTI Act Section cited (Section 8/Section 24) |
| First_Appeal_Filed | String | Yes / No |
| First_Appeal_Date | Date | Date first appeal was filed |
| First_Appeal_Outcome | String | Upheld / Partially Upheld / Rejected |
| Second_Appeal_CIC | String | Yes / No — escalated to CIC |
| Second_Appeal_Outcome | String | CIC ruling |
| Is_Repeat_Filer | String | Yes / No |
| Compliant_Response | String | ENGINEERED — Yes/No/Pending (30-day rule) |
| Breach_Days | Int | ENGINEERED — Days past deadline (0 if on-time) |
| Filing_YearMonth | String | ENGINEERED — e.g. "2019-03" |
| Financial_Year | String | ENGINEERED — e.g. "FY 2023-24" |
| Proactive_Disclosure_Candidate | String | ENGINEERED — Yes/No flag for Section 4 |

# Appendix B — SQL Query Index

All 15 queries are available in full in **sql/rti_analysis_queries.sql** and **notebooks/03_sql_analysis.ipynb**. They are designed for SQLite but run unchanged on PostgreSQL and MySQL.

| Query | Theme | Description |
|---|---|---|
| Q1 | Volume & Growth | Year-wise RTI volume and YoY growth rate |
| Q2 | Seasonal | Monthly filing pattern — peak month identification |
| Q3 | Department | Department-wise volume, share & compliance rate |
| Q4 | Categories | Top RTI categories — most frequently filed |
| Q5 | Compliance Trend | Year-wise compliance rate (30-day deadline) |
| Q6 | Disposal | Disposal type breakdown — overall & year-wise |
| Q7 | Rejection | Section-wise rejection reason analysis |
| Q8 | Appeal Funnel | RTI → First Appeal → CIC escalation funnel |
| Q9 | Appeal Outcomes | First appeal outcome distribution |
| Q10 | Breach Rate | Departments with highest non-compliance rate |
| Q11 | PIO Performance | PIO workload & individual compliance performance |
| Q12 | Geography | State-wise geographic distribution (Top 10) |
| Q13 | Digital Adoption | Online vs offline filing trend by year |
| Q14 | COVID Impact | 2019 vs 2020 vs 2021 comparison |
| Q15 | Sec 4 Disclosure | Proactive disclosure candidates under Section 4 |